A Single Drone. Forty Sensors. Thirty-Five Minutes.

Energy-Efficient UAV Path Planning in IoT Networks: A Deep Reinforcement Learning Aided Approach

Atilade Gabriel Oke

Final Presentation, May 2026

EEEN30330 Individual 3rd-Year Project
Department of Electrical & Electronic Engineering
The University of Manchester

The scene

A remote sensor field.

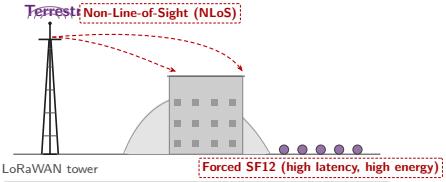
Forty sensors.

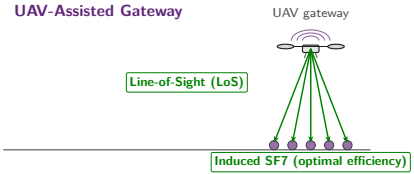
One drone.

35 minutes of battery.

The drone is the only gateway. Every uncollected sensor is lost data.

And it is also a protocol-controller, not just a data-mule.





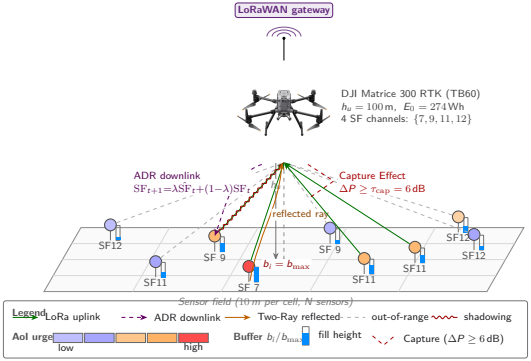
Two coupled problems, one control input

Physical

Limited battery, large grid, partial coverage. Drone position determines link budget.

Protocol: *mobility-blind ADR*

LoRaWAN's Adaptive Data Rate is engineered for static gateways. Under a moving UAV it triggers the **SF monoculture**: every sensor converges on SF12, intra-SF collisions collapse throughput.



UAV position is the shared control input.

Gap: no prior work addresses physical *and* protocol jointly on a mobile gateway.

Five things in 15 minutes

1. **The problem** two coupled constraints, no closed form
2. **My approach** DQN trained across 16 conditions simultaneously
3. **What I found** performance collapses at scale; a greedy heuristic beats it
4. **Why, and the fix** representational ceiling diagnosed; permutation-invariant architecture
5. **The evidence** NDR $+333\%$, energy $\times 12$, all $p < 10^{-4}$, 50 seeds

Problem formulation

Objective: maximise total data collected

$$\max_{\pi} \sum_{t=0}^{T} \sum_{i=1}^{N} D_{i,t}$$

Subject to:

$$\mathrm{RSSI}_{i,t}(p_t) \geq \mathrm{RSSI}_{\min}(SF_{i,t}) \quad \forall i$$

$$E_{t+1} = E_t - P(a_t)\Delta t \geq 0 \quad \forall t$$

$$\sum_{i \in \mathcal{N}} \mathbf{1}[SF_{i,t}{=}k] \leq 1 \quad \forall k \in \{7, \ldots, 12\}$$

$$p_{t+1} = p_t + v(a_t)\,\Delta t$$

$D_{i,t}$: bytes from sensor i at step t. π: control policy. E_t: battery. $SF_{i,t}$: spreading factor.

Three forces act simultaneously:

Battery

$$E_{t+1} = E_t - P(a_t)\Delta t$$

$P_{\text{hover}} > P_{\text{flight}}$; hovering costs more.

Coverage

$$P_r \propto d^{-4}$$

Two-Ray: halve distance $\Rightarrow$ 16$\times$ power.

Protocol lag

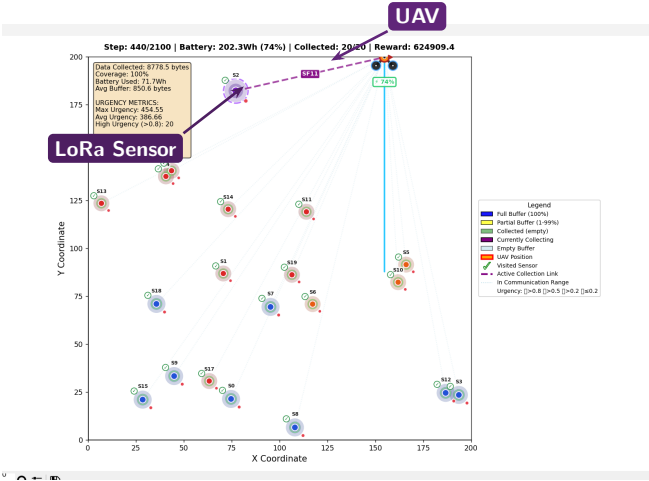
$$\bar{R}_t = \lambda R_t + (1-\lambda)\bar{R}_{t-1}$$

EMA-ADR, $\lambda{=}0.1$; SF lags $\sim$10 s.

Non-convex. Non-stationary. No closed form.

$\Rightarrow$ trained a **Deep Q-Network** to find π.

Experimental setup: the simulator



Environment at a glance

- **Grid**: 200×200 m, UAV at 100 m altitude
- **LoRa sensors (S0–S19)**: ring colour = buffer fill (blue = full, green = empty)
- **UAV** (orange): collecting from S2 at SF11; dashed line = active link
- **Urgency rings**: AoI priority — darker ⇒ data loss imminent
- **Dotted circle**: comm. range — 17 of 20 reachable
- **Episode**: 2100 steps ≈ 35 min

The agent: DQN with domain randomisation

Observation (4-frame stack)

UAV position · battery · per-sensor {buffer fill, urgency, link quality}

Domain randomisation

4 grid sizes $\times$ 4 sensor counts $= 16$ conditions simultaneously. One policy, deployed everywhere.

5 discrete actions

N · S · E · W · Hover, chosen every second. Flat MLP $[512, 512, 256]$, ReLU, Adam.

At small scale: parity

Three comparators:

Nearest Greedy	Move to closest buffered sensor. No protocol knowledge.
SF-Aware Greedy	Adds protocol awareness; picks highest-throughput sensor.
TSP Oracle	Optimal offline route. Routing upper bound.

50 seeds per condition, 5 grid sizes $(100\times100, N{=}10) \rightarrow (500\times500, N{=}50)$. Welch's t-test.

$$100\%$$

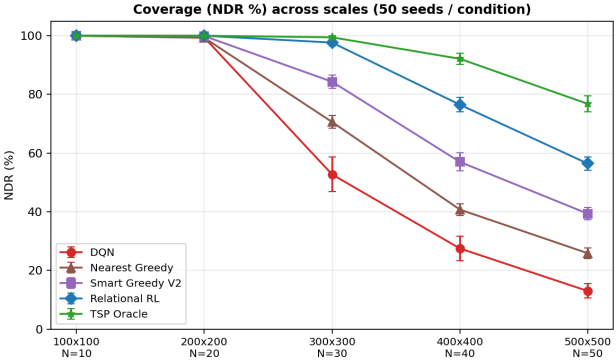
DQN NDR at $(100\times100,\ N{=}10)$

$$99.4\%$$

DQN NDR at $(200\times200,\ N{=}20)$

All agents at parity. The DQN is working.

At scale: collapse



13.0%

DQN NDR at $(500 \times 500,\ N{=}50)$

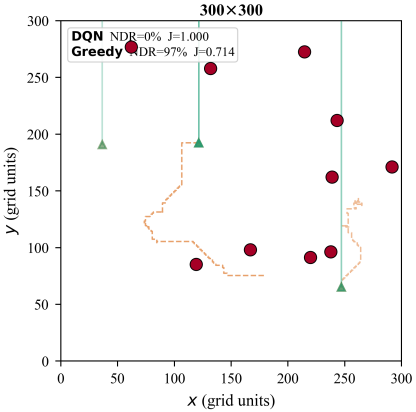
Protocol-blind Nearest Greedy: 25.9%**.**

DQN 52.7% vs Nearest 70.6% at $(300, 30)$.

DQN 13.0% vs Nearest 25.9% at $(500, 50)$.

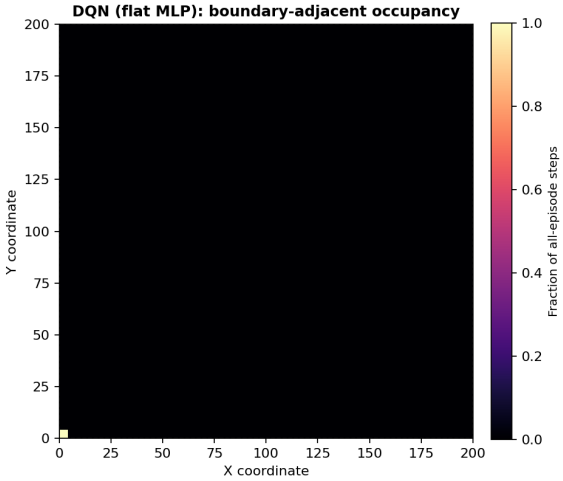
The bottleneck is not the protocol.

What is the DQN actually doing?



DQN: NDR = 0% — never leaves its column. Greedy: NDR = 97% — structured sweep, reaches almost every sensor. The drone is physically capable of moving. It doesn't.

Diagnostic: 100% boundary-cell occupancy



20 deterministic episodes,
$(200 \times 200, N{=}20)$

$$100\%$$

boundary-cell occupancy

Median 0 wall collisions:
boundary avoidance *was* learned.

But the interior is never entered.

Reward penalises both boundary hits and starvation. The agent still patrols the perimeter.

Physically capable. Architecturally constrained.

The cause: representational ceiling

Not reward. Not training. Not physics.

The flat MLP consumes $[\mathrm{uav},\ s_1,\ s_2,\ \ldots,\ s_N]$ as a single concatenated vector.

The problem

Sensor order is part of the representation. Swapping sensor 1 and sensor 2 $\Rightarrow$ the policy sees a completely different input.

At small scales: memorise the fixed layout. At large scales: cannot generalise.

The consequence

Perimeter patrol is the cheapest *stable* trajectory the network can express under this encoding.

The architecture cannot treat the sensor set as a set.

The fix: an architectural change

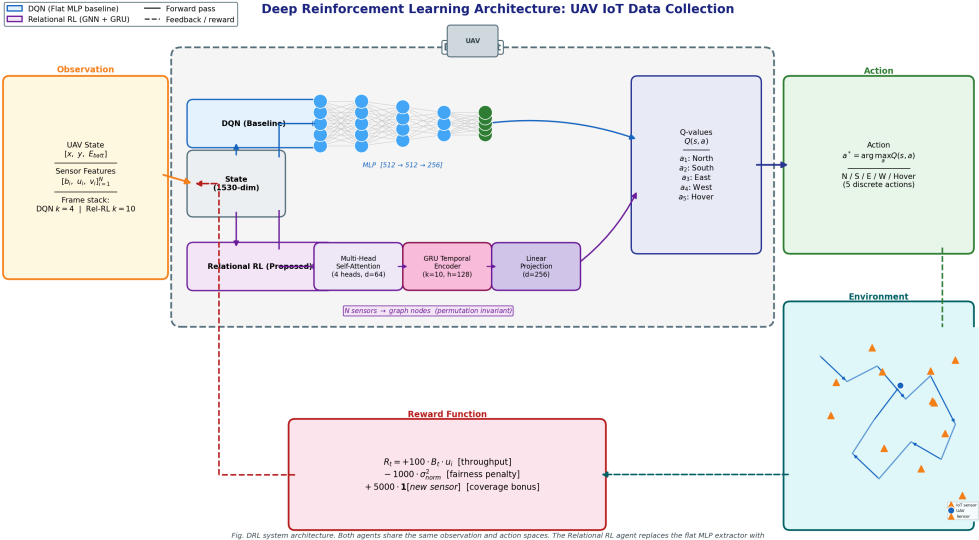
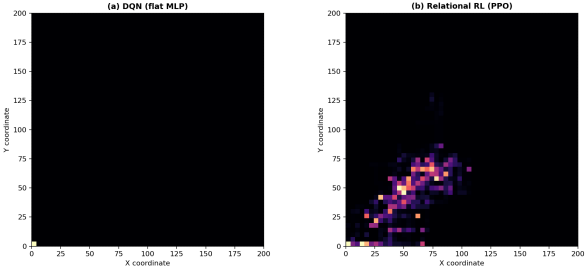


Fig. DRL system architecture. Both agents share the same observation and action spaces. The Relational RL agent replaces the flat MLP extractor with sensor self-attention (spatial GNN) and a GRU temporal encoder, enabling permutation-invariant reasoning over a variable number of IoT sensors.

Diagnostic resolved



Boundary-cell occupancy

$$100\% \rightarrow 21.5\%$$

Wall collisions

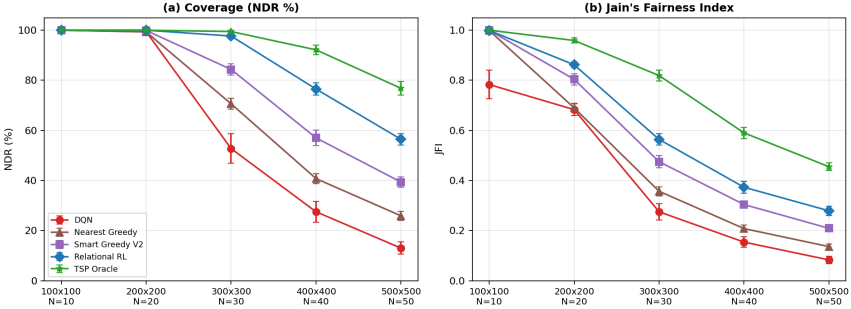
$$21.8 \rightarrow 0$$

across all 20 seeds

Left: DQN stuck in one corner. Right: Relational RL ranges across the grid.

Root cause confirmed. Fix

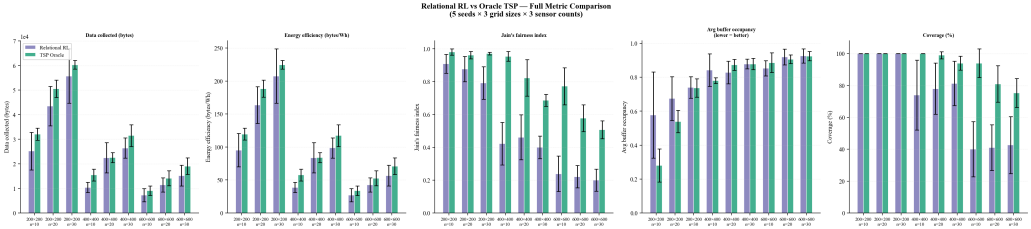
Results: scalability restored



Metric at $(500\times500, N{=}50)$	DQN	Relational RL	
NDR (% sensors visited)	13.0%	$\mathbf{56.4}\%$	$(+333\%)$
Jain's fairness index	0.08	$\mathbf{0.28}$	
Energy efficiency (B/Wh)	9.96	$\mathbf{121.9}$	$(\times12)$

All differences: $p < 10^{-4}$, Welch's t-test, 50 seeds per condition. $t = -26.4$ at $(500, 50)$.

Results: how close to optimal?



TSP Oracle = omniscient upper bound (pre-computed tour, no mid-flight adaptation). 5 seeds × 3 grid sizes × 3 sensor counts. Error bars = std.

Throughput & efficiency

-14% overall; gap narrows at trained-scale grids (200×200).

Coverage

73% vs 94% NDR — driven by 600×600 OOD degradation.

Fairness

Oracle's uniform tour ≠ smarter.
Rel. RL weights **urgent** sensors.

Contributions and what's next

Contributions

1. Open Gymnasium env: Two-Ray + ADR + Capture Effect + shadowing, all coupled
2. Domain-randomised competence-gated curriculum
3. 50-seed benchmark against Greedy and TSP Oracle
4. **Architectural diagnostic:** flat-MLP perimeter heuristic as a representational ceiling
5. **Permutation-invariant fix:** pathology resolved, scalability curve shifted by $+333\%$ NDR

The diagnostic-and-fix pairing is the principal claim.

Future directions

- **Attention pooling** over sensor set (close the 20-point gap to TSP Oracle)
- **Multi-UAV** coordination (QMIX, MAPPO) (beyond single-drone battery envelope)
- **Hardware-in-the-loop** validation (sim-to-real robustness already tested)

Thank you.

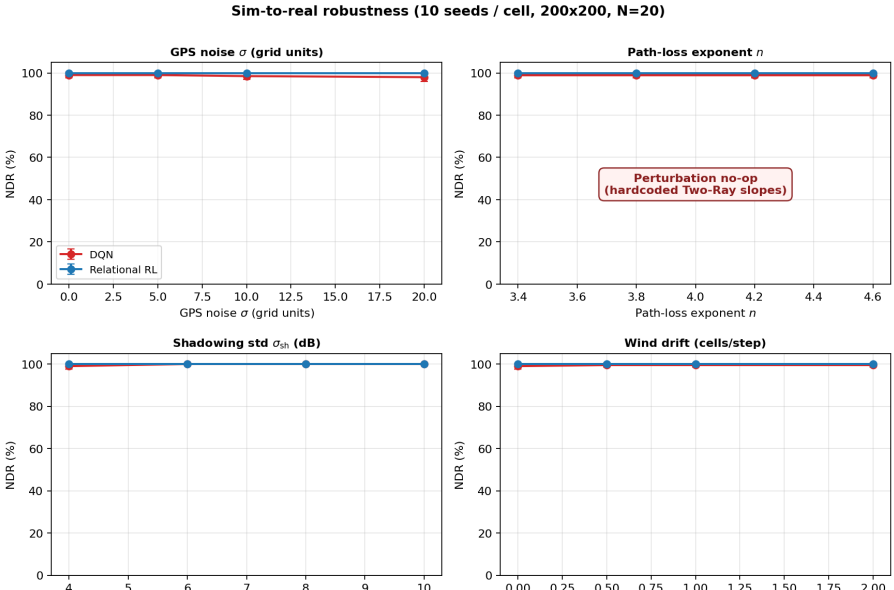
Questions?

Backup: ablation study

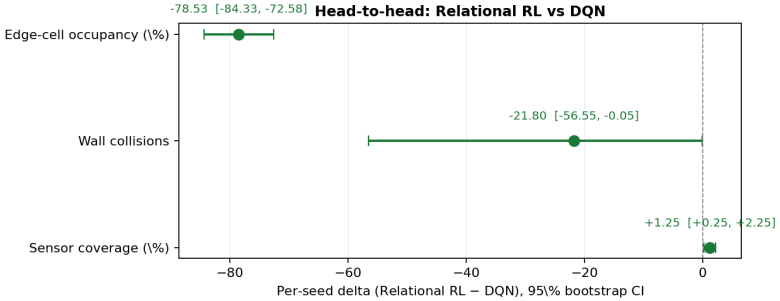
Metric	Full mean	Ablation mean	Cohen's d	p
Cumulative reward ($\times 10^5$)	8.24	2.88	+1.43	0.00012
NDR	0.186	0.138	+0.54	0.098
Jain's fairness	0.288	0.177	+1.23	0.00052
Gini (lower is better)	0.730	0.833	−1.32	0.00023
Energy efficiency (B/Wh)	51.4	35.8	+0.63	0.055

Ablation = no domain randomisation (A4). 20 seeds, Welch's t-test. Every large-effect component ($|d|>0.8$) reaches $p<0.001$.

Backup: sim-to-real robustness

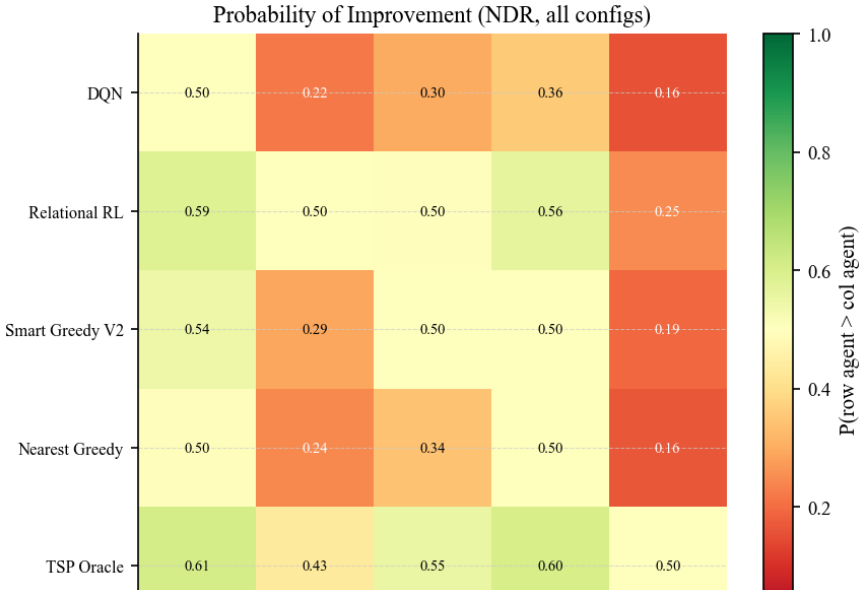


Backup: head-to-head forest plot

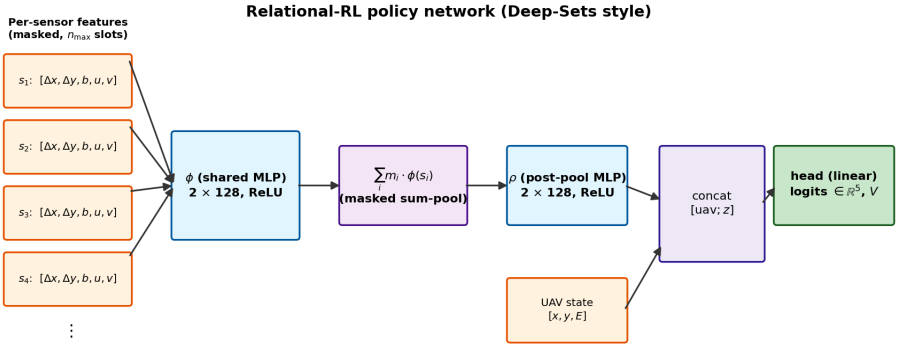


Seed-matched deltas (Relational RL − DQN) across three diagnostic metrics, 95% bootstrap CIs (n=10,000). All intervals exclude zero.

Backup: probability of improvement (50 seeds)



Backup: Relational RL network detail



Permutation invariance: reordering $\{s_i\}$ leaves the masked sum unchanged.

$$z = \rho\left(\sum_{i=1}^{n_{\max}} m_i\, \phi(s_i)\right)$$

ϕ: shared MLP per sensor Σ: masked sum-pool ρ: post-pool MLP